Xu Pan M.Sc.

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LinkedIn GitHub ResearchGate Scholar

Experiences

Nanyang Technological University (NTU) Singapore | 2026.08 - present
Research Associate | *School of Electrical and Electronic Engineering, College of Engineering*

- Under the supervision of Prof. Ziwei Wang, at Perception and Embodied Intelligence (PINE) Lab
- VLA Reinforcement Learning: Investigating policy optimization and exploration for flow-based VLAs.
- Dual-System Architectures: Exploring asynchronous VLAs with learned latent-state updates for efficient control.
- Adaptive Robot Learning: Developing feedback-conditioned in-context learning and flow guidance for manipulation.

Agency for Science, Technology and Research (A*STAR) Singapore | 2025.08 - 2026.06
Research Assistant | *Centre for Frontier AI Research (CFAR), Institute of High Performance Computing (IHPC)*

- Pioneered spatially-aware reinforcement learning adaptation for flow-based VLA models, preserving implicit spatial inductive bias through fused geometric prior and preventing degradation under distribution shifts.
- Designed geometry-aligned dense rewards and spatially-conditioned exploration, enabling stable RL fine-tuning, superior zero-shot spatial generalization, and robust transferable behaviors in complex manipulation scenes.

Baidu International Technology (Shenzhen) Co., Ltd. Shenzhen, China | 2024.12 - 2025.03
Research Intern | *Foundation Public Technology Group, International Technology R&D Department*

- Developed DiT-based framework for controllable, arbitrarily long video generation via key-frame conditioning.
- Improved the visual fidelity of rendered text in image generation by designing specialized attention mechanisms.
- Contributed to the **SynClub** AI chat app through assessment of frontier multimodal generative models and optimization of text-to-image/video modules, delivering improved performance and higher-quality outputs.

Education

Wuhan University | **State Key Lab. LIESMARS** Wuhan, China | 2023.09 - 2026.06
M.Sc. in Cartography and Geographic Information Systems

- GPA: 3.54/4.00, Average Score: 89.40/100, IELTS: 7.0(6.0)
- Research area: Computer Vision and 3D Reconstruction

Wuhan University | **School of Remote Sensing and Information Engineering** Wuhan, China | 2019.09 - 2023.06
B.Eng. in Remote Sensing Science and Technology

- GPA: 3.75/4.00, Average Score: 88.90/100, Recommended Exemption

Publications

1. **Xu Pan**, Zhenglin Wan, Xingrui Yu, Xianwei Zheng, Youkai Ke, Ming Sun, Rui Wang, Ziwei Wang, Ivor Tsang. **SA-VLA: Spatially-Aware Reinforcement Learning for Flow-Matching Vision-Language-Action Models**. *IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops (CVPRW)*, 2026.
2. **Xu Pan**, Zimin Xia, Xianwei Zheng. **Scale-aware Co-visible Region Detection for Image Matching**. *ISPRS Journal of Photogrammetry and Remote Sensing [JCR Q1, IF 12.2]*, 2025. `https://doi.org/10.1016/j.isprsjprs.2025.08.015`
3. **Xu Pan**, Zhenglin Wan, Xingrui Yu, Xianwei Zheng, Youkai Ke, Ming Sun, Rui Wang, Ziwei Wang, Ivor Tsang. **SFM: Spatially-Aware Flow Matching for Embodied Reinforcement Learning**. *Under Review*, 2026. Available at `https://arxiv.org/abs/2602.00743`.
4. **Xu Pan**, Zhen Pang, Qiyuan Ma, Mingyue Dong, Wei Ji, Xianwei Zheng. **SAMatcher: Dense Co-Visibility Modeling via Cross-View Fusion for Scale-Imbalance Image Matching**. *Under Review at IEEE Transactions on Circuits and Systems for Video Technology*, 2026. Available at `https://arxiv.org/abs/2606.03406`
5. Ming Sun, Rui Wang, Xingrui Yu, Lihua Jing, Hangyu Du, Zhenglin Wan, **Xu Pan**, Ivor Tsang. **Towards Backdoor-Based Ownership Verification for Vision-Language-Action Models**. *Under Review at IEEE Transactions on Pattern Analysis and Machine Intelligence*, 2026. Available at `https://arxiv.org/abs/2605.09005`

6. Jiashen Huang, Yu Jia, **Xu Pan. Trusting AI in China: Institutional Trust and the Domestic Advantage of DeepSeek over ChatGPT**. *Under Review*, 2026. Available at `https://arxiv.org/abs/2606.01228`.

Projects

Intelligent Interpretation Platform for Remote Sensing Images Based on Baidu PaddlePaddle 2022.03 - 2022.08
`TwSphinx54/I2RSI`

Web-based Global Natural Disaster Assessment System 2022.05 - 2022.09
`TwSphinx54/GNDAS`

Academic Activities

2026 Vision and Learning Seminar (VALSE), China Society of Image and Graphics (CSIG) | Wuhan, China 2026.05.08 - 05.10

2025 Annual Conference on Photogrammetry and Remote Sensing, Chinese Society for Geodesy, Photogrammetry and Cartography (CSGPC) | Kunming, China 2025.07.28 - 07.30

Awards and Honors

- **Outstanding Graduating Graduate Student**, Wuhan University 2026.05
- **Outstanding Graduate Student**, Wuhan University 2025.11
- **Graduate Academic Excellence Scholarship**, Wuhan University 2025.11
- **Graduate Academic Excellence Scholarship**, Wuhan University 2024.11
- **Outstanding Student Leader**, Wuhan University 2024.10
- **Outstanding Student Club Leader**, Wuhan University 2024.08
- **Active Contributor to Social Activities**, School of Remote Sensing and Information Engineering 2023.05
- **Wuhan University Class C Scholarship**, Wuhan University 2022.10
- **National Second Prize**, China Software Cup College Student Software Design Competition 2022.08
- **Honorable Mention**, Mathematical Contest In Modeling 2022.03
- **Third Prize**, Asia and Pacific Mathematical Contest in Modeling 2021.12
- **First Prize in Hubei Division**, China Undergraduate Mathematical Contest in Modeling 2021.10
- **Outstanding Student**, Wuhan University 2021.09
- **Bronze Medal**, China Collegiate Algorithm Design & Programming Challenge Contest 2021.03
- **Second Prize in Final**, Translation & Interpreting Contest of Hubei Province 2020.05

Skills

Languages: Mandarin, English (IELTS 7.0)

Programming: `Python`, `C++`, `MATLAB`, `HTML`, `CSS`, `JavaScript`, `C#`, `SQL`, `R`

Soft Skills: Leadership, Project Management, Event Organization, Public Speaking, Design & Editing

Certificates

System Integration Project Management Engineer
Qualification Certificate of Computer and Software Technology Proficiency, MOHRSS & MIIT

Certificate of Level 4 in Network Engineer
National Computer Rank Examination, NEEA

Certificate of Level 3 in Network Technology
National Computer Rank Examination, NEEA

Certificate of Level 3 in Database Technology
National Computer Rank Examination, NEEA

Academic Services

Reviewer	2027 International Joint Conference on Neural Networks (IJCNN'27)
Reviewer	10th Annual Conference on Robot Learning (CoRL'26)
Member	ISPRS Student Consortium (ISPRS SC)
Volunteer	2022/2023 International Graduate Workshop on Geo-Informatics
	The 21st China Ocean Color Conference

Leadership

Chairman	The 22nd LIESMARS Graduate Student Union
Director	GeoScience Café Operation Center